

PROXIMITY SENSOR SWITCH

For Wireless Control Of Electric Loads

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Workers' safety is paramount in the public as well as private sector. Yet, electric shock is one of the major safety concerns in the electric power industry. According to a report, 30 people get killed every day due to electrocution, and many more get injured in the electric power industry. The majority of these deaths occur among the unskilled and semi-skilled workers.

High-pressure working conditions and minimal rest time cause these unfortunate incidents. During short resting periods in small restrooms, sometimes, people in a hurry touch loose and bare electric wire connections with their bare feet to turn on fans, AC, coolers, or any other electric equipment. At times, this results in a severe electric shock, leading to death or injury. Moreover, control rooms in the power industry usually have many wires, and most of the time, in real-life scenarios, fan or cooler connections are loose, posing a threat to the well-being of the workforce working in such control rooms.

To eliminate these real-time threats in the

power system industry, here is a system based on an IR sensor module to control the load, especially fans, coolers, or AC units. This low-cost system operates in wireless mode, eliminating the need for physical touch to turn the load on and off.



Fig. 1: EFY Lab's prototype

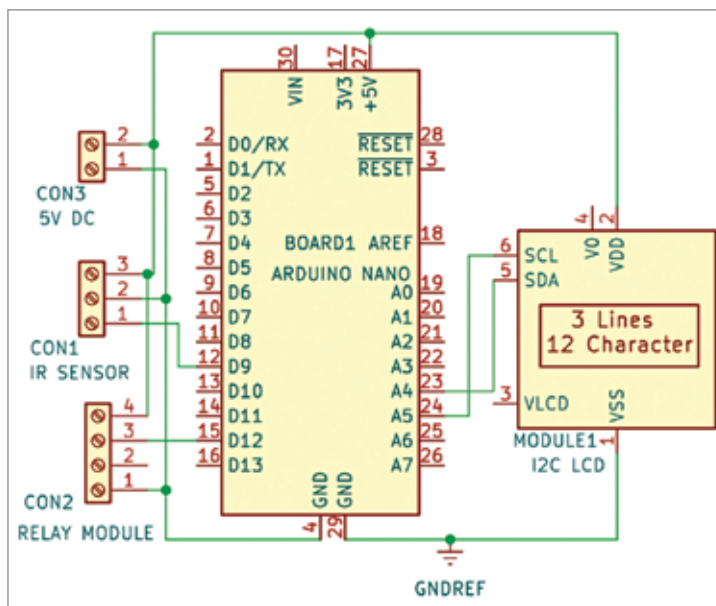


Fig. 2: Circuit diagram

Fig. 1 shows EFY Lab's prototype for this load control project.

Circuit and working

The circuit diagram of the wireless load control unit is shown in Fig.

2. This design is based on Arduino Nano board and built around an I2C LCD (module1). It uses an Arduino Nano (board1), an IR sensor module, a 5V relay module, jumper wires, a breadboard, and a few other components.

The output signal pin of the IR sensor module (connected across CON1) is linked to D9 (pin 12) of Arduino Nano, which serves as an input. The relay module connected to D12 (pin 15) of Arduino Nano acts as an output pin. 230V AC is connected using the relay's contacts to control the on/off operation of AC loads like fans, AC units, and coolers.

The Arduino Nano is a micro-sized and breadboard-compatible board based on micro-processor ATmega328. The Arduino board can be powered via the Mini-B USB connection. The power source is automatically selected